

Cross-Cultural Social Acceptability Classification via Cultural Context and Norm Retrieval

Jessica Hung, Suhas Julapalli, Kuo-Peng Huang

COM SCI 263

UCLA

Abstract

Social acceptability judgments are not universal: the same behavior can be acceptable, unacceptable, or context-dependent depending on cultural norms, relationship roles, and situational details. We study whether language models can adapt their social judgments when given explicit cultural context. We construct a 120-example cross-cultural social acceptability classification dataset with three labels: *acceptable*, *not acceptable*, and *context-dependent*. Each example includes a scenario, a culture or social-context label, and a short cultural-context explanation. We evaluate several GPT-based prompting strategies, including scenario-only classification, cultural-context prompting, retrieval-augmented generation (RAG), and Social-Chem-101 retrieval with explicit intermediate norm findings. We separately analyze scalar acceptability rating as an ablation. Our results show that cultural context is the largest driver of improvement: GPT-4o mini improves from 57.50% accuracy in the scenario-only baseline to 79.16% when cultural context is included. The reported GPT-4o mini RAG setup combines cultural context with retrieved norm evidence and achieves 87.50% accuracy. A newer GPT-4o + Social-Chem-101 RAG experiment achieves 89.17% accuracy while producing inspectable intermediate norm claims, making errors easier to analyze. These findings suggest that culturally grounded social acceptability classification benefits not only from stronger models, but also from prompt formats that represent ambiguity and retrieval methods that surface relevant social norms.

1 Introduction

Social acceptability classification asks whether a behavior, statement, or interaction is socially appropriate in a given context. This task is difficult because social norms are culturally situated. For example, calling a boss by their first name may

be ordinary in an informal U.S. workplace but inappropriate in a setting with stronger hierarchy or formality expectations. Similarly, whether skipping a family dinner is acceptable may depend on cultural expectations around family obligation, personal boundaries, frequency, and explanation.

Large language models often produce fluent social judgments, but fluency does not guarantee cultural adaptability. A model may apply a generic or majority-culture norm even when the scenario provides a different cultural context. This is especially risky for conversational AI systems that give advice, evaluate behavior, or mediate social interactions across communities.

In this project, we investigate the following research question: **Can language models adapt social acceptability judgments when the same behavior is paired with different cultural norms?**

We focus on three goals. First, we build a small but controlled dataset of culturally sensitive social scenarios. Second, we evaluate whether explicit cultural context improves model predictions. Third, we test whether richer prompting and retrieval-augmented generation (RAG) can help models apply cultural information more consistently, especially for ambiguous cases.

Compared with our earlier project plan, we remove DeBERTa from the main experimental story. Although DeBERTa was useful as a preliminary zero-shot NLI baseline, we did not train or fine-tune it on our task. Therefore, its results are less informative for our final research question than the GPT, RAG, and open-source LLM experiments.

Code Availability. The complete implementation, including prompts, retrieval pipelines, evaluation scripts, and experiment configurations, is available at: <https://github.com/06huange/Social-Acceptability-Classification>.

2 Related Work

Prior work has shown that language models and benchmarks often encode social assumptions from their data sources. NLPositionality (Santy et al., 2023) argues that datasets reflect the positionality of their creators and annotators. GeoMLAMA (Yin et al., 2022) shows that commonsense knowledge varies across geographic and cultural contexts. Recent cultural and safety evaluation benchmarks, including SALAD-Bench (Li et al., 2024) and NORMAD (Rao et al., 2025), similarly highlight that model behavior may not generalize uniformly across social settings.

Our project also draws on retrieval-augmented generation (Lewis et al., 2020), which improves model behavior by providing task-relevant external knowledge at inference time. Instead of retrieving factual passages, we retrieve social norms and rules of thumb. We also use Social-Chem-101 (Forbes et al., 2020), a large dataset of social and moral rules of thumb, as a broad norm source for one of our RAG experiments.

3 Dataset

We constructed a dataset of 120 manually curated social acceptability examples. Each example contains a scenario or interaction, a culture or social-context label, a cultural-context explanation, and one gold label from `acceptable`, `not acceptable`, or `context-dependent`.

The dataset covers domains such as family expectations, privacy and consent, workplace hierarchy, communication style, politeness, friendship, fairness, authority, care/harm, and commitment. Many scenarios are intentionally paired across cultural contexts so that the same behavior may receive different labels depending on the norm description. This design tests whether a model can use cultural guidance rather than assigning one generic judgment to all variants.

Label	Count
<code>acceptable</code>	23
<code>context-dependent</code>	47
<code>not acceptable</code>	50
Total	120

Table 1: Dataset label distribution.

The `context-dependent` label is important because many social judgments depend on missing or variable factors, such as relationship closeness, consent, public vs. private setting, hierarchy,

severity, frequency, or prior agreement. This label also prevents forcing all cases into a binary acceptable/unacceptable distinction.

4 Methods

4.1 Scenario-Only Baseline

The first baseline gives the model only the scenario text and asks it to classify the behavior into one of the three labels. The prompt explicitly instructs the model not to use cultural context unless it is included in the situation. This tests whether the model can solve the task using generic social knowledge alone.

4.2 Cultural-Context Prompting

The next setup appends the dataset’s culture label and cultural-context explanation to the prompt. This tests whether explicit cultural guidance improves classification. For example, instead of only seeing “I called my boss by their first name,” the model may also see a context such as “Japan, formal-hierarchy norm” with an explanation that formal address and seniority are important.

4.3 Retrieval-Augmented Generation (RAG)

We construct a controlled RAG database from the dataset’s cultural-context fields. This database excludes gold labels and prior model predictions to avoid answer leakage. The retrieved chunks summarize cultural norms, scenario patterns, and conditions under which a behavior is acceptable, unacceptable, or context-dependent. We test whether adding retrieved norm evidence improves the model’s ability to apply the cultural context consistently.

4.4 Social-Chem-101 RAG with Intermediate Findings

Finally, we build a large RAG database from Social-Chem-101. We filter out low-quality rules of thumb, aggregate repeated annotations by rule ID, and preserve fields such as the rule of thumb, action, moral foundation, agreement, cultural pressure, legality, and inferred project-category hints. The full corpus contains 285,514 chunks, and the project-matched subset contains 265,743 chunks.

Unlike the dataset-derived RAG database, Social-Chem-101 is not culture-specific. We therefore treat it as broad social and moral norm evidence, while the dataset’s cultural-context explanation remains the primary source for culture-specific

judgment.

For this experiment, GPT-4o receives the scenario, culture label, cultural-context explanation, and top retrieved Social-Chem-101 norms. The model must output structured JSON including intermediate RAG-grounded findings, whether each norm applies to the scenario, whether the evidence supports a label or only background context, a relevance judgment for the retrieved norms, and the final label. This format helps us audit whether errors are caused by bad retrieval, irrelevant norms, or incorrect model reasoning.

5 Experiments and Results

We report the most meaningful experiments rather than every ablation. The early DeBERTa baseline is omitted from the main table because it was not trained or fine-tuned for this task.

The results show three main trends. First, scenario-only classification is weak because the model often applies a generic social judgment to culturally different cases. Second, adding explicit cultural context produces the largest single improvement. Third, retrieval can further improve accuracy when the retrieved evidence is well matched to the scenario and culture. We analyze the effect of rating-based prompting separately in the ablation section.

Because the dataset contains only 120 examples, these results should be interpreted as preliminary rather than definitive. Relatively small changes in the number of correctly classified examples can produce noticeable differences in overall accuracy. Larger-scale evaluation will be necessary to determine whether the observed trends generalize.

The GPT-4o mini context + RAG setup achieves 87.50% accuracy. This suggests that a compact, task-aligned norm database can improve culturally grounded classification compared with using cultural context alone. However, the Social-Chem-101 RAG reasoning experiment is still valuable because it exposes intermediate norm claims and makes the model’s reasoning easier to inspect.

6 Ablation Study

6.1 Effect of Rating-Based Prompting

We analyze the 1–10 scalar acceptability rating as an ablation rather than as a separate main method. In this prompt format, the model first assigns an acceptability score from 1 to 10, where low scores indicate clear social unacceptability,

high scores indicate clear acceptability, and middle scores indicate ambiguity or missing context. Scores are then mapped to labels: 1–3 = not acceptable, 4–7 = context-dependent, and 8–10 = acceptable.

The comparison shows that adding a rating step improves the context-only setup from 79.16% to 83.33%. This suggests that allowing the model to express degree before committing to a discrete label is helpful for borderline and context-dependent cases. Adding RAG produces a larger improvement, reaching 87.50%, because retrieved norm evidence gives the model additional scenario-specific guidance.

7 Analysis

7.1 Why Cultural Context Helps

The scenario-only baseline often fails because many examples are underdetermined without cultural information. For instance, a model may judge skipping a family gathering as acceptable based on individual autonomy, but in another cultural context family gatherings may carry stronger relational obligations. Adding the cultural-context explanation gives the model a specific norm to condition on, reducing reliance on default assumptions.

7.2 Why Retrieval Helps Sometimes

Retrieval helps when the retrieved norms match the scenario’s key social variables. In the RAG setup, retrieved chunks often include culture-specific conditions such as hierarchy, family involvement, privacy, reciprocity, or public embarrassment. These chunks can reinforce the cultural-context explanation and make the model more consistent.

However, retrieval can hurt when the retrieved evidence is too generic or points to a broad moral rule that does not capture the culture-specific nuance. This was visible in the Social-Chem-101 experiment. For example, retrieved norms such as “posting someone’s picture without consent is wrong” may be valid in general, but they can overpower a cultural-context explanation where consent, audience, or cultural expectations make the case more nuanced.

7.3 Social-Chem-101 Reasoning Results

The GPT-4o + Social-Chem-101 RAG reasoning experiment achieved 89.17% accuracy. Its per-label performance was 82.61% for acceptable, 85.11% for context-dependent, and 96.00%

Experiment	Input / Method	Accuracy
Scenario-only baseline	GPT-4o mini, scenario only	69/120 = 57.50%
Context baseline	GPT-4o mini, scenario + culture/context	95/120 = 79.16%
Context baseline	Llama 3 7B, scenario + culture/context	73/120 = 60.83%
Context baseline	Qwen 3 8B, scenario + culture/context	105/120 = 87.5%
Social-Chem RAG reasoning	Qwen 3 8B + thinking, Social-Chem-101 RAG + intermediate findings	103/120 = 85.83%
Context + RAG	GPT-4o mini, scenario + culture/context + retrieved norms	105/120 = 87.50%
Social-Chem RAG reasoning	GPT-4o, Social-Chem-101 RAG + intermediate findings	107/120 = 89.17%
Social-Chem RAG reasoning	Llama 3 7B, Social-Chem-101 RAG + intermediate findings	96/120 = 80.0%
Social-Chem RAG reasoning	Qwen 3 8B, Social-Chem-101 RAG + intermediate findings	98/120 = 81.67%
Social-Chem RAG reasoning	Qwen 3 8B + thinking, Social-Chem-101 RAG + intermediate findings	101/120 = 84.17%

Table 2: Selected experiment results. DeBERTa is excluded from the final comparison because it was not trained or fine-tuned for this task.

Setting	Input / Method	Accuracy
Context baseline	Scenario + culture/context	95/120 = 79.16%
Context + rating	Scenario + culture/context + 1–10 rating	100/120 = 83.33%
Context + RAG	Scenario + culture/context + retrieved norms	105/120 = 87.50%

Table 3: Ablation results for rating-based prompting and RAG.

for not acceptable. The model was strongest on clearly unacceptable cases, but it sometimes over-applied broad negative norms to cases labeled acceptable or context-dependent. The structured output helped reveal this pattern: in some privacy cases, the model retrieved valid general norms about consent, but the final prediction became too strict relative to the dataset’s cultural-context explanation.

7.4 Error Categories

We observed three recurring error types. First, broad rules such as “breaking agreements is bad” or “posting without consent is wrong” are usually valid, but may be too strong for cases where the cultural context makes the behavior acceptable or conditional. Second, some prompts overuse or under-use context-dependent; they either classify ambiguous cases as clearly unacceptable because a retrieved norm sounds negative, or classify clearly unacceptable cases as context-dependent because missing details are imaginable. Third, RAG helps only when the retrieved examples align with the scenario’s core norm; if retrieval finds a related but not exact norm, the model may reason from the wrong social variable.

8 Open-Source Model Experiments

We tested two open source models, Llama 3 7B and Qwen 3 8B for a subset of the tests with GPT-4o to offer further results comparisons. We tested both models with the baseline cultural context prompt and the Social-Chem-101 RAG enhanced prompt. As well, we further tested Qwen 3 8B with thinking

enabled on both prompts. All tests were done on an Apple Macbook M4 Pro. Tests that did not utilize thinking were ran on MPS, whereas the Qwen with thinking tests were run on MLX with a conversion of the same model weights.

The results show distinct patterns between Llama and Qwen. Llama achieved a test accuracy of 0.6083 on the baseline cultural context prompt and an accuracy of 0.8 using the RAG-enhanced prompt. Qwen, however, showed the inverse pattern. Qwen achieved an accuracy of .875 with the baseline cultural context prompt, which dropped to .8167 with the enhanced prompt. Interestingly, Qwen 3 8B achieved higher accuracy than GPT-4o mini under the cultural-context prompt despite having substantially fewer parameters. One possible explanation is that the benchmark’s modest size makes results sensitive to prompt formatting and example composition. Additional evaluation on larger datasets would be needed before drawing broader conclusions about relative model capability. A similar trend was found with thinking enabled, with Qwen achieving an accuracy of 0.8583 with the basic prompt and 0.84 with the enhanced prompt.

Llama 3 7B showed the same trend as GPT-4o, limited initial performance that was enhanced by the introduction of RAG decision rules. Qwen, however, did not show a clear pattern. Looking to table 4, we can see that Qwen’s precision and recall stats without thinking are initially evenly distributed. Introducing the RAG-enhanced prompt reduced the precision for acceptable and the recall for context dependent. This implies the model is

Class	Basic Context			Social-Chem-101 Rag		
	P	R	F1	P	R	F1
Acceptable	0.78	0.91	0.84	0.83	0.83	0.83
Not Acceptable	0.84	0.98	0.91	0.86	0.86	0.86
Context-Dependent	0.94	0.70	0.80	0.83	0.83	0.83
Macro Avg	0.86	0.87	0.85	0.84	0.84	0.84
Weighted Avg	0.87	0.86	0.85	0.84	0.84	0.84

Table 4: Classification Statistics for Qwen 3 7B + thinking.

now over attributing acceptable situations to context dependent with the decision rules. With thinking enabled, the model initially over attributes acceptable situations to being context dependent, and evens out the statistics when provided with the RAG-enhanced prompt. In both cases, the RAG-enhanced prompt reduced overall accuracy, but with respect to precision and recall the prompt improved the overall distribution when thinking was applied, while damaging it without thinking. Based on the results, the utilization of decision rules with Qwen is primarily useful for reducing acceptability mis-attribution when thinking is enabled. For Llama, however, the decision rules are immediately and strongly useful.

9 Limitations

Several limitations should be considered when interpreting these results.

First, the dataset contains only 120 examples, which limits statistical power and may make performance estimates sensitive to individual examples. Second, the dataset was manually constructed and labeled by the project team. Although labels were assigned through discussion and consensus, we did not collect annotations from multiple independent annotators or measure inter-annotator agreement.

Third, the dataset covers only a limited set of cultures, social contexts, and interaction types. The benchmark should therefore be viewed as an exploratory evaluation rather than a comprehensive measure of cultural adaptability. Fourth, Social-Chem-101 primarily provides broad social and moral norms rather than explicitly culture-specific knowledge. Retrieved norms may therefore fail to capture important cultural nuances.

Finally, our experiments focus on prompting and retrieval-based adaptation rather than model fine-tuning. Future work may find different performance trends when models are trained directly on culturally grounded datasets.

10 Future Work

Several directions could strengthen this line of research. The most immediate priority is expanding the dataset to include more cultures, regions, languages, and social situations. Future versions should incorporate multiple independent annotators and report inter-annotator agreement to better characterize ambiguity in social judgments.

Another promising direction is the development of culture-specific retrieval corpora rather than relying primarily on broad social norm resources such as Social-Chem-101. More targeted retrieval sources may improve performance on culturally nuanced scenarios.

Future work could also investigate supervised fine-tuning of open-source models, multilingual evaluation, and agent-based retrieval systems that dynamically gather cultural evidence before making social acceptability judgments.

11 Conclusion

Our experiments show that cross-cultural social acceptability classification is not simply a generic commonsense task. Models perform much better when they receive explicit cultural context, and additional gains are possible through task-aligned retrieval and rating-based prompting. The reported GPT-4o mini RAG configuration achieved 87.50% accuracy using scenario text, cultural context, and retrieved norm evidence. A GPT-4o Social-Chem-101 RAG experiment achieved 89.17% accuracy while producing intermediate norm findings, which improved interpretability and accuracy.

The main lesson is that culturally grounded reasoning requires more than generic commonsense knowledge. Effective social acceptability classification depends on a model’s ability to appropriately balance cultural context with broader social norms and rules of thumb, particularly in ambiguous situations.

For reproducibility, all code and experimental artifacts are publicly available at:

[https://github.com/06huange/
Social-Acceptability-Classification](https://github.com/06huange/Social-Acceptability-Classification).

References

- Maxwell Forbes et al. 2020. Social Chemistry 101: Learning to Reason about Social and Moral Norms.
- Patrick Lewis et al. 2020. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks.
- Sebastian Santy et al. 2023. NLPositionality: Characterizing Design Biases of Datasets and Models.
- Da Yin et al. 2022. GeoMLAMA: Geo-Diverse Commonsense Probing on Multilingual Pre-Trained Language Models.
- Lei Li et al. 2024. SALAD-Bench: A Hierarchical and Comprehensive Safety Benchmark for Large Language Models.
- Rao et al. 2025. NormAd: A Framework for Measuring the Cultural Adaptability of Large Language Models.